

SAFETY STOCK & ROP REPORT

Operations Analytics-1: Inventory Policy Report

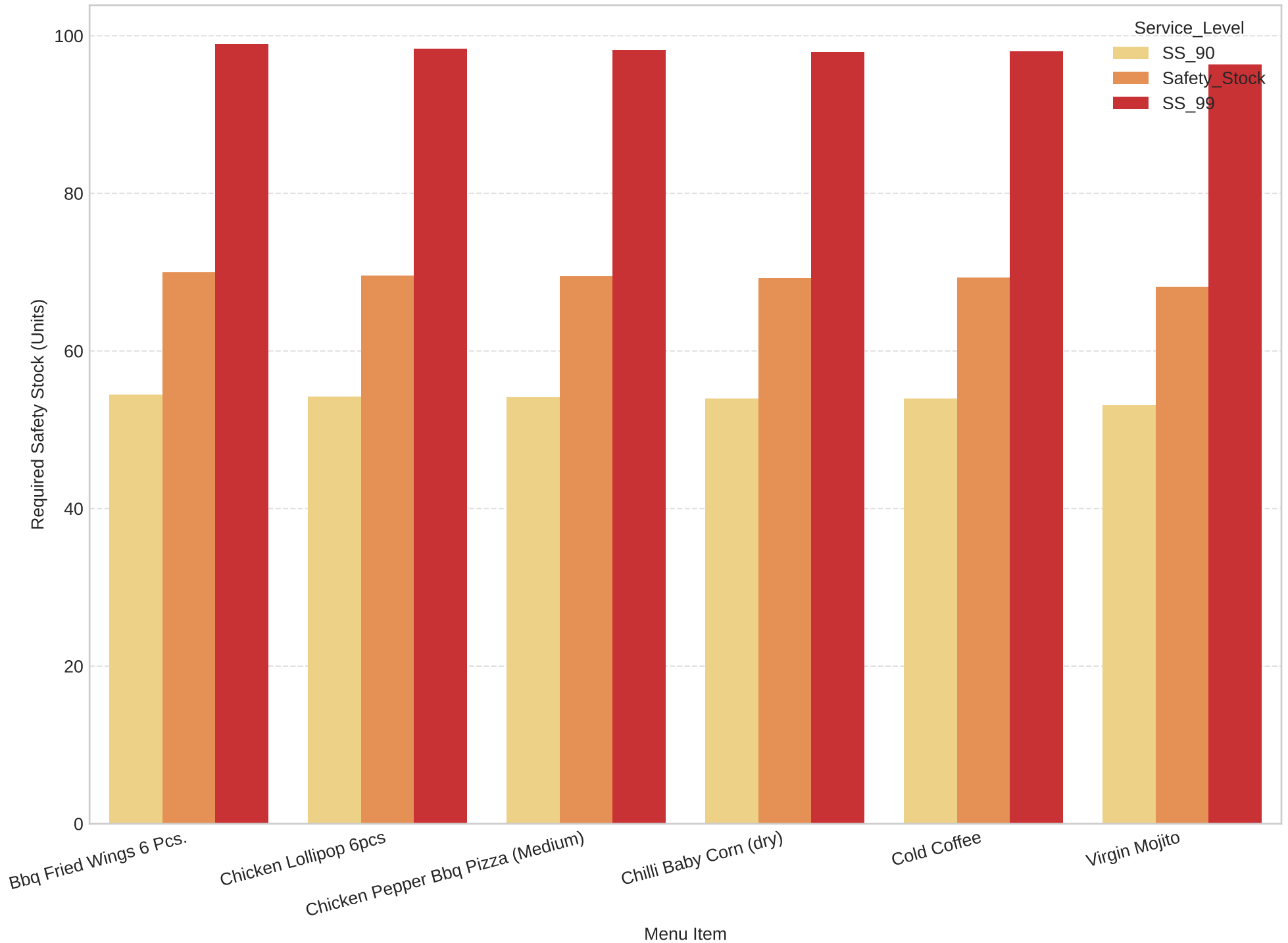
Algorithm: Probabilistic Safety Stock & Reorder Point Optimization

Objective: Calculate exact mathematical buffers to guarantee a 95% service level against demand volatility and supply chain lead-time delays.

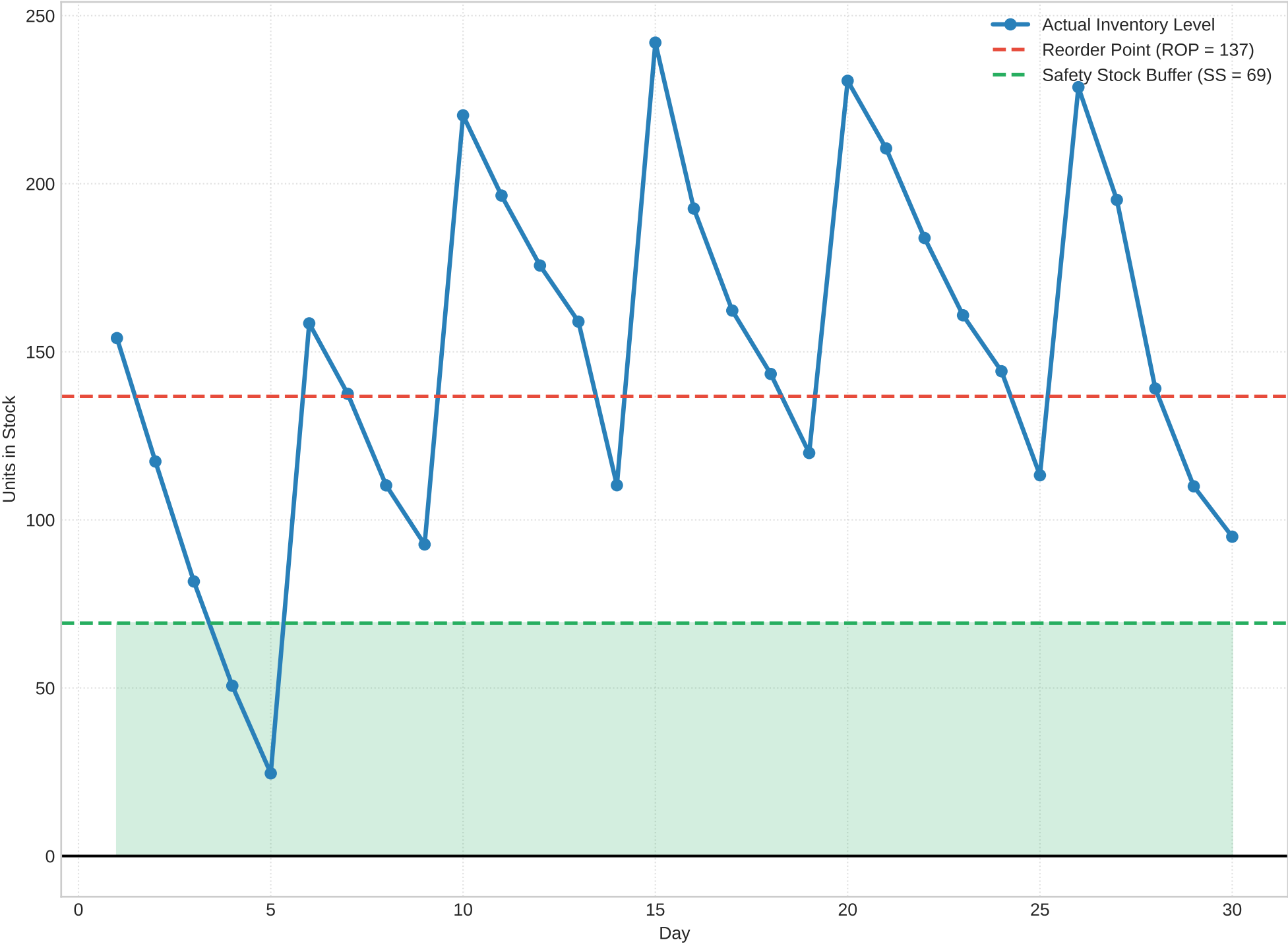
Master Inventory Parameters (95% Service Level)

Item_Name	Mean_Daily_Demand	Std_Dev_Demand	Safety_Stock	Reorder_Point (ROP)
Bbq Fried Wings 6 Pcs.	29.8	8.0	69.9	138.1
Chicken Lollipop 6pcs	29.8	7.3	69.5	137.8
Chicken Pepper Bbq Pizza (Me	29.6	7.8	69.4	137.2
Chilli Baby Corn (dry)	29.5	7.9	69.2	136.7
Cold Coffee	29.5	8.0	69.3	136.7
Virgin Mojito	28.9	8.3	68.1	134.2

Sensitivity Analysis: Capital Required for Higher Service Levels



Digital Twin: 30-Day Inventory Cycle Simulation (Cold Coffee)



EXECUTIVE INVENTORY POLICY

Inventory Policy Implementation Strategy:

The Digital Twin simulation visually proves the necessity of the Reorder Point (ROP). As inventory drains over time, crossing the red dashed line ($ROP = 137$) acts as the mathematical trigger to place a new order.

The green shaded region represents the Safety Stock ($SS = 69$). Notice how the inventory line occasionally dips into the green zone but stays above zero.

Strategic Action:

By standardizing these triggers in our ERP/WMS system, we mathematically protect the kitchen from unexpected demand spikes or supplier delays. Pushing for a 99% service level requires disproportionately higher capital tied up in inventory (as seen in the Sensitivity Analysis), hence 95% remains our optimal operational threshold.